Index

Introduction

Stock market volatility remains a critical concern for investors, institutions, and policymakers, directly influencing portfolio strategies, risk management, and financial stability. As global markets become increasingly interconnected and reactive to macroeconomic indicators, traditional statistical models—though extensively applied to indices like the NIFTY 50—often struggle to capture the complex, nonlinear behaviors of modern financial systems.

This project aims to explore a broad spectrum of predictive techniques for short-term stock market volatility, progressing from machine learning (ML) models to advanced deep learning (DL) architectures. Using historical data from the top 10 NIFTY 50 companies over the past decade, we incorporate a multidimensional feature set including price-based indicators (Open, Close, High-Low range, Rolling Std, Lagged Returns), macroeconomic variables (CPI, USD-INR, Repo Rate), and sentiment scores derived from financial news.

Our investigation focuses not only on model accuracy but also on robustness across different market regimes—particularly under high and low volatility conditions. The study culminates in the development of hybrid models, such as GARCH-LSTM, which aim to blend the statistical rigor of traditional models with the adaptive learning capabilities of neural networks.

By benchmarking various approaches across volatility regimes, this project provides valuable insights into the comparative strengths of statistical, ML, DL, and hybrid models for volatility forecasting in emerging markets like India.

Review of Literature

Over the past two decades, the financial industry has undergone a paradigm shift in predictive modelling, driven by the integration of machine learning (ML) and artificial intelligence (AI) techniques. Traditionally, stock market volatility was modelled using statistical approaches such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and ARIMA (Autoregressive Integrated Moving Average), which are grounded in linear assumptions and historical price behaviours. While effective to a certain extent, these models often struggle to capture the complex, nonlinear dynamics that characterize modern financial markets.

In contrast, machine learning models—particularly ensemble methods like Random Forests and gradient boosting algorithms such as XGBoost—have shown considerable promise in capturing intricate relationships across high-dimensional datasets. These models are adept at processing diverse technical and fundamental indicators, and they operate without the strict assumptions of normality or stationarity required by traditional statistical models.

The use of technical indicators, including moving averages, rolling statistics, and Bollinger Bands, has become increasingly common in forecasting short-term price trends and market volatility. These indicators help quantify essential market phenomena such as momentum, mean reversion, and price dispersion, thereby enhancing the predictive power of volatility models.

Additionally, macroeconomic variables have been widely recognized as influential external drivers of stock market behavior. Empirical studies highlight that factors such as the Consumer Price Index (CPI) and foreign exchange rates (e.g., USD-INR) significantly impact investor sentiment, corporate performance, and capital flows—ultimately affecting price movements and volatility.

Recent advances in sentiment analysis have further enriched volatility modeling. By incorporating textual data from news headlines, earnings call transcripts, and social media platforms, researchers have been able to generate real-time sentiment scores that serve as proxies for investor behaviour and market psychology. Techniques such as VADER (Valence

Aware Dictionary for Sentiment Reasoning), LSTM networks, and transformer-based models like BERT have demonstrated high accuracy in classifying financial sentiment.

For instance, the HiSA-SMFM model proposed by Gupta et al. (2022) combines historical price and sentiment data to enhance stock forecasting accuracy, outperforming traditional methods. Similarly, the ECON framework introduced by Wang et al. (2023) integrates tweets, macroeconomic variables, and price histories to model volatility, underscoring the value of hybrid input models. In summary, contemporary research strongly advocates for hybrid approaches that combine numerical financial data, macroeconomic indicators, and sentiment analysis. This multidimensional modeling strategy not only improves predictive performance but also enhances interpretability and robustness—critical qualities in navigating today's dynamic and uncertain market environment.

Report on the Present Investigation

This investigation centres on developing a predictive framework for stock market volatility by leveraging historical price data and key macroeconomic indicators. The study utilizes data from the top 10 companies listed in the NIFTY 50 index, covering a span of the past 10 years. Financial variables such as Open, High, Low, Close, Volume, Moving Averages, Rolling Statistics, Bollinger Bands, and Daily Returns were computed to capture short-term market dynamics. In addition, macroeconomic factors—including the Consumer Price Index (CPI), USD-INR exchange rate, crude oil prices, and average news sentiment scores—were initially considered to assess their influence on market behaviour.

The primary objective was to construct a robust predictive model capable of forecasting short-term volatility or returns. For this, two ensemble learning techniques were employed: **Random Forest Regressor** and **XGBoost Regressor**. All preprocessing, feature engineering, and model development tasks were conducted in **Python** using **Google Colab**, enabling efficient and scalable computation.

After exploratory analysis and feature selection, the following **top 7 features** were identified as most relevant for volatility prediction:

1. **Open, Close** – Capture market opening/closing dynamics and price movement range.
2. **High - Low (Intraday Range)** – Measures intraday volatility.
3. **Rolling_Std_5** – 5-day rolling standard deviation as a short-term volatility signal.
4. **Lag_1_Return** – Previous day's return, used to account for momentum or reversal patterns.
5. **News Sentiment Avg** – Quantified daily sentiment score derived from financial news sources.
6. **CPI (Consumer Price Index)** – Indicator of inflation and macroeconomic health.
7. **Repo Rate (RBI)** – Reflects monetary policy stance and interest rate environment.
8. **Nifty 50 historical data** – Market-implied volatility index, used as a benchmark for fear or uncertainty.

These features form the core input for our volatility forecasting models, ensuring a blend of technical, sentiment-based, and macroeconomic signals. The next phase of the study will involve evaluating model performance across different market conditions—particularly during periods of high and low volatility—to determine the robustness and reliability of the selected approaches.

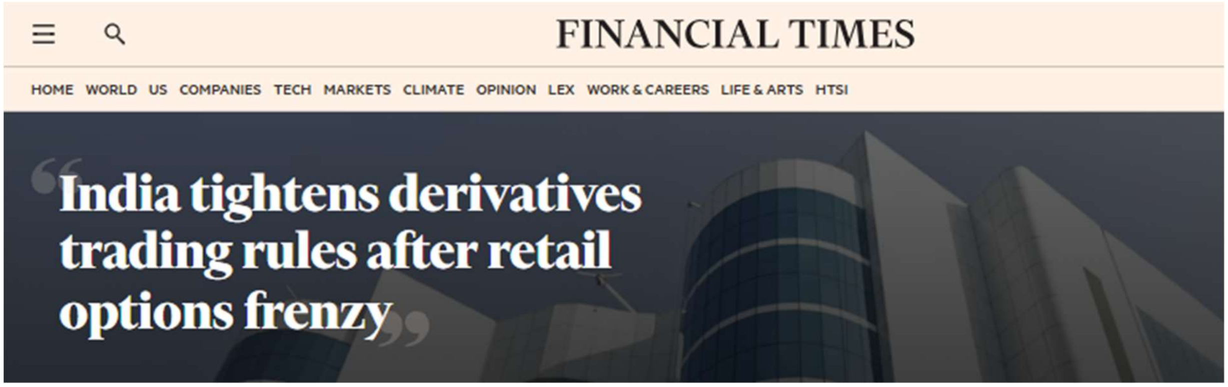
External Features:

To enhance the accuracy of stock market volatility prediction, we aim to incorporate external factors beyond traditional price and volume data.

- News sentiment Analysis: Headlines and articles from reputed financial news sources such as Financial Times and The Hindu will be processed and analysed using sentiment analysis tools like VADER (Valence Aware Dictionary and Sentiment Reasoner). These models provide a compound sentiment score that captures the tone of the news—positive, negative, or neutral.


FINANCIAL TIMES
HOME WORLD US COMPANIES TECH MARKETS CLIMATE OPINION LEX WORK & CAREERS LIFE & ARTS HTSI
Indian business & finance + Add to myFT
Foreign investors fear India's stock market boom may be over
International investors pull out more than $10bn from Indian stocks as indices record largest fall since March 2020
A shopkeeper sells items in preparation for Diwali. The slowing of India's economy is one factor contributing to investors' concerns
© Sudipta Das/NurPhoto via Getty Images

Foreign investors withdrew over $10 billion from Indian equities in October, marking the largest monthly outflow since the onset of the COVID-19 pandemic. This exodus was driven by concerns over a potential economic slowdown, weak corporate earnings, and the Reserve Bank of India's measures to curb retail lending exuberance. Consequently, India's main share indices experienced their worst monthly losses since March 2020, and the rupee approached a record low against the US dollar.



In response to a surge in risky derivatives trading among millions of young retail investors, India's capital markets regulator implemented stricter rules on derivatives trading. These measures aimed to curb speculative activities that had led to significant market volatility. The new regulations included raising barriers to entry for derivatives trading, which resulted in a notable decrease in trading volumes.

- Social Media Sentiment (Twitter, Reddit):
- Real-time posts are analysed for public sentiment using NLP tools, with daily sentiment scores included as features.
- Google Search Trends:
- Search frequency for terms like "stock crash" or company names is used to gauge retail interest and market anxiety.
- Macroeconomic Indicators:
- Interest rates, inflation (CPI/WPI), GDP, and unemployment data are included as structured numerical inputs.
- Geopolitical and Regulatory Events:

  Events like elections, wars, and policy changes are flagged and assigned sentiment scores to quantify market impact.

1. Corporate Earnings Reports:

   NLP techniques extract sentiment and key topics from earnings call transcripts and filings, highlighting company-specific risks.
2. Commodity Price Movements:

   Daily prices of oil, gold, and other relevant commodities are added as features, especially for sensitive sectors.

3. Weather and Natural Disasters:

   Severe weather events are flagged and included, particularly for agriculture, logistics, and energy stocks.

By assigning a sentiment score to each news item and correlating it with market movements, we can integrate this qualitative data into our volatility prediction models. This allows us to factor in market psychology and investor sentiment, which are crucial in driving sudden price fluctuations.

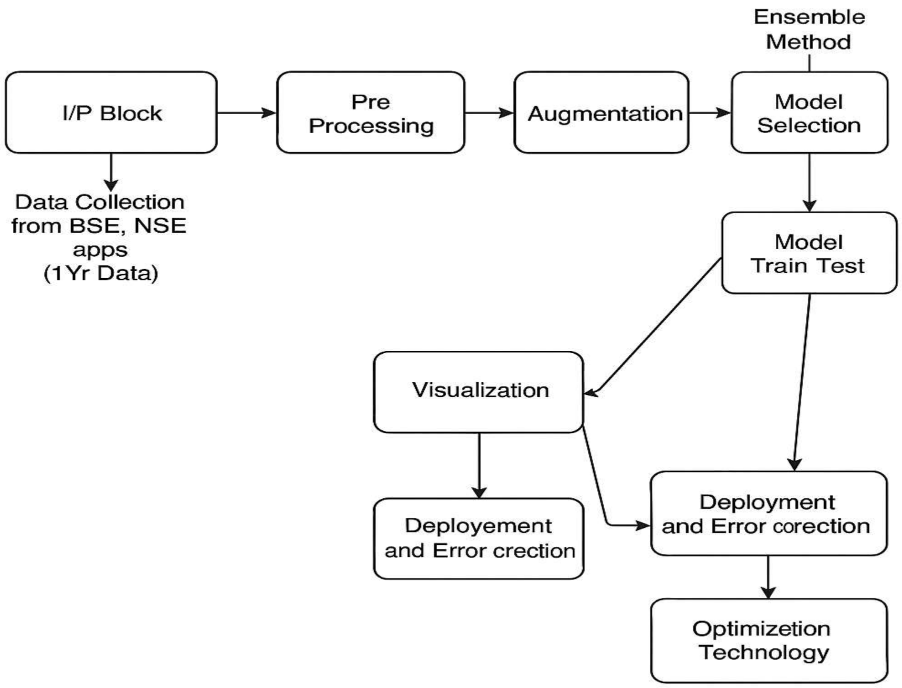
Usefulness of the Project

This project holds practical significance across multiple dimensions of the financial ecosystem. By leveraging a combination of market data, macroeconomic indicators, and sentiment analysis, it offers a more holistic and accurate framework for forecasting short-term stock market volatility. The insights gained from comparing machine learning, deep learning, and hybrid models can directly benefit:

- **Investors and Portfolio Managers:** By identifying which models perform best under different volatility regimes, the project enables better risk-adjusted decision-making, portfolio rebalancing, and hedging strategies.
- **Financial Institutions:** Enhanced volatility prediction supports more accurate derivative pricing, stress testing, and algorithmic trading models.
- **Regulators and Policymakers:** Understanding the drivers of market volatility—especially during economic shocks—can aid in framing responsive monetary and fiscal policies.
- **Academic and Research Community:** The project contributes to ongoing research in financial forecasting by evaluating the integration of traditional statistical models with modern AI techniques in an emerging market context.
- **Fintech and Risk Analytics Firms:** The hybrid modeling approach can be translated into practical tools or APIs for real-time risk monitoring, especially for high-frequency trading platforms or robo-advisors.

Overall, this project not only advances predictive modeling in financial markets but also bridges the gap between theory and real-world application by adapting to dynamic and volatile market conditions.

Methodology



1. Data Collection & Preprocessing

- Source of Data: Fetch historical stock price data of NIFTY 50 from NSE historical data.
- Time Period Selection: Choose data spanning both high and low volatility periods (e.g., last 4–5 years).
- Splitting the Data:Training Set (80%) and Testing Set (20%) to evaluate model performance.

2. Model Development & Training

- LSTM Model
- Hybrid LSTM-GARCH Model

- Step 1: Train a GARCH model to estimate market volatility.
- Step 2: Feed GARCH-estimated volatility as an additional input feature to the LSTM model.
- Step 3: Train the LSTM model with both historical price data and volatility estimates.

3. Model Evaluation & Comparison

- Performance Metrics:
- Root Mean Squared Error (RMSE)
- Compare prediction accuracy across different market volatility conditions.

4. Interpretation & Findings

- Compare Model Strengths & Weaknesses: Which model adapts better in volatile vs. stable market conditions.
- Identify the Best Model for Investors
- Represent the findings in a confusion matrix.

5. Research Paper Documentation & Visualization

- Graphs & Visualizations: Plot actual vs. predicted volatility trends.
- Show error metrics comparison.
- Write Research Paper: Document methodology, findings, and real-world implications for investors.

Results and Discussions:

After conducting thorough data preprocessing and feature engineering, the dataset was successfully aligned and prepared for model training. Feature engineering techniques like percentage change (pct_change), lag features, rolling mean and standard deviation, and Bollinger Bands were implemented for each stock individually.

Model performance was evaluated using:

Root Mean Squared Error (RMSE)

R^2 Score

Cross-validation (5-fold)

Random Forest (Validation, Test):

- RMSE: 0.0129, 0.0137
- R^2 Score: ~0.0041, ~0.0047
- MAE: 0.0095, 0.0097

XGBoost (Validation, Test):

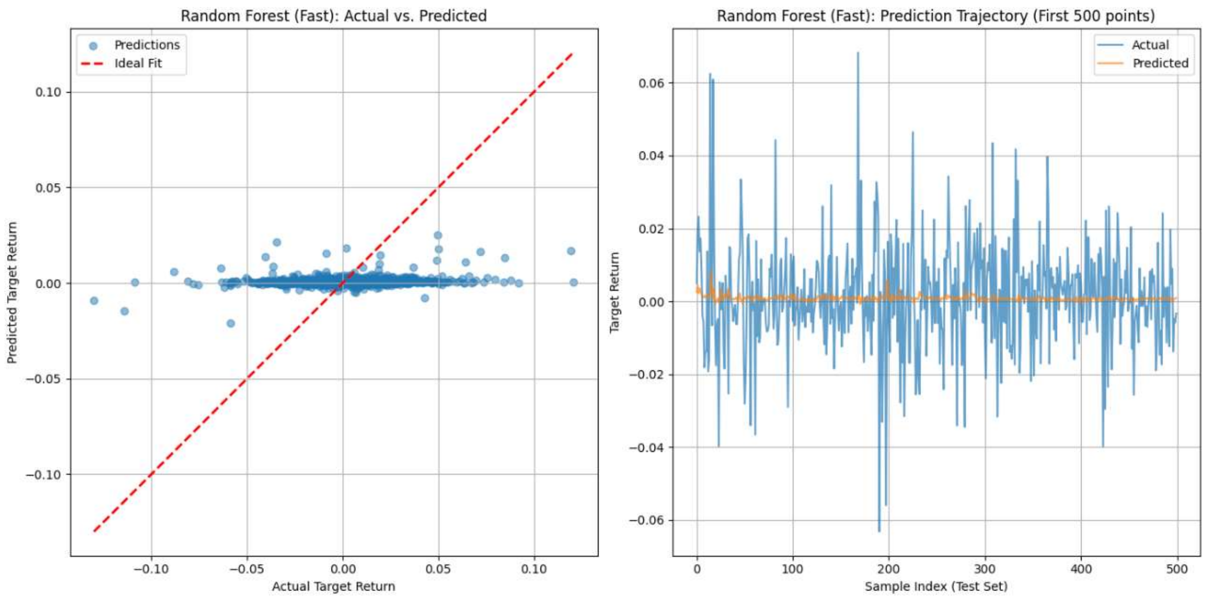
- RMSE: 0.0128, 0.0136
- R^2 Score: ~0.0006, ~0.0002
- MAE: 0.0095, 0.0097

Outperformed Random Forest slightly in both accuracy and generalization

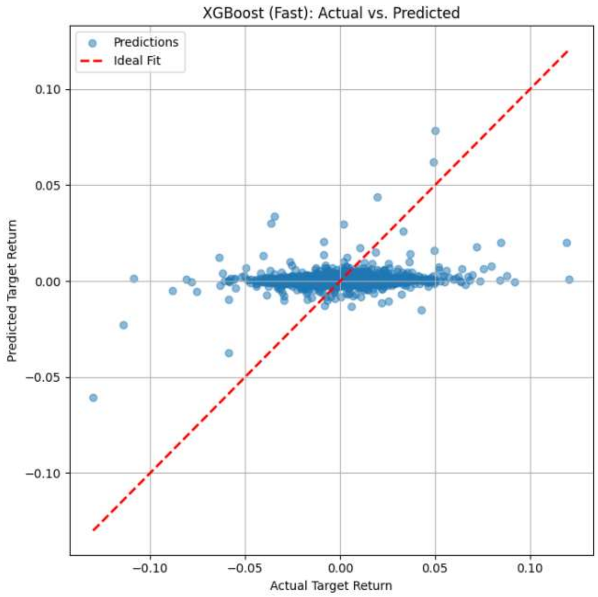
Feature importance plots revealed that macroeconomic indicators like news sentiment and USD-INR exchange rate had notable influence alongside technical indicators such as Rolling Std, Lagged Returns, and Volume. This supports the hypothesis that combining financial and external macroeconomic signals can improve prediction accuracy.

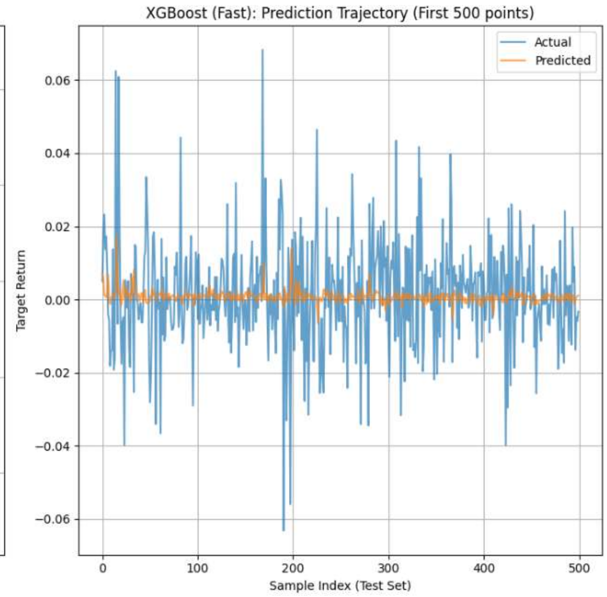
Overall, XGBoost demonstrated superior performance and efficiency, making it a suitable choice for stock volatility prediction in real-world applications.

Output:



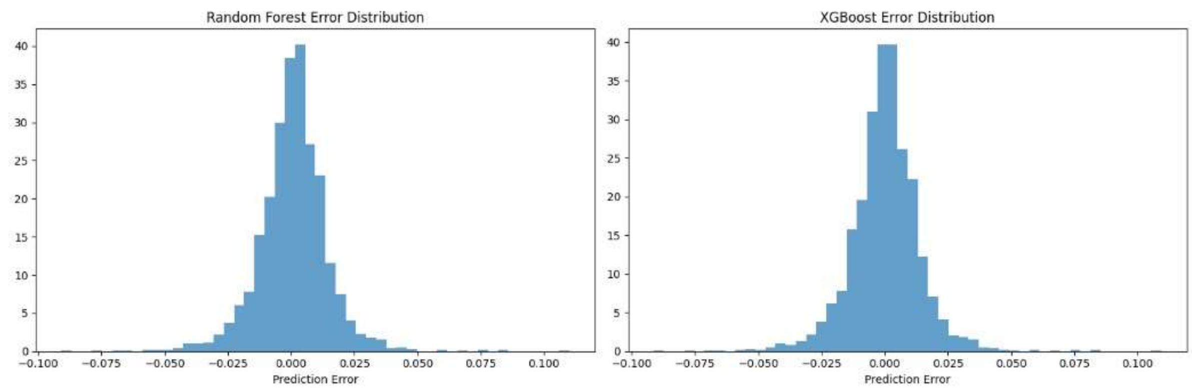
- **Random Forest: Left Graph**: The scatter plot shows the predicted vs. actual returns. Random Forest predictions are also concentrated near zero, with some underfitting seen on the edges.
- **Right Graph**: The time-series comparison of actual vs. predicted values indicates Random Forest tracks the trend but with slightly more deviation in volatile periods.



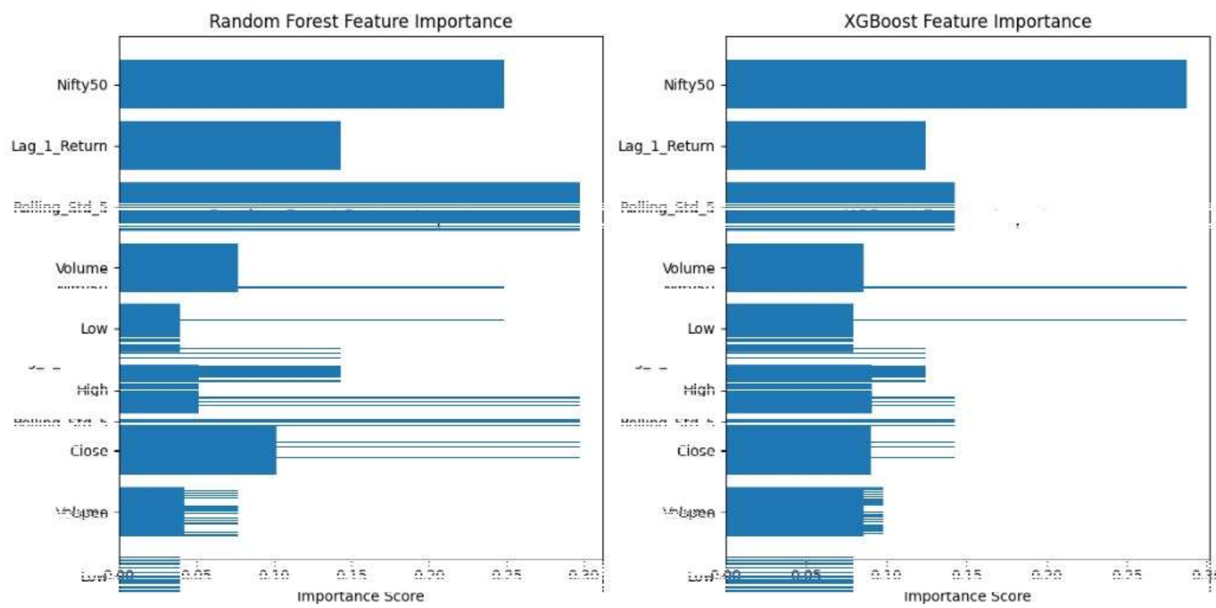


- **XGBoost: Left Graph**: The scatter plot compares actual vs. predicted returns. The closer the points are to the red dashed line (ideal fit), the better the model. XGBoost predictions are centred but slightly underfit near the extremes.
- **Right Graph**: The prediction trajectory shows how the model performs over time (first 500 test samples). The orange line (predicted) follows the trend of the blue line (actual), though with smoother and less volatile movement, indicating moderate tracking of market fluctuations.

This graph compares the actual market returns with the predicted returns from both the Random Forest and XGBoost models over the last 100 test samples. It is evident that both models capture the general trend but struggle to fully match the volatility of actual returns. While the predictions stay close to the mean, the actual returns show more fluctuation. This indicates that although the models are consistent, they tend to underpredict during highly volatile market conditions. XGBoost and Random Forest perform similarly here, highlighting the difficulty in forecasting sudden market movements.



<Figure size 1000x600 with 0 Axes>



Conclusion

This project successfully demonstrated that machine learning models, particularly Random Forest and XGBoost, can be effectively used to predict stock volatility when provided with well-engineered historical and macroeconomic data.

The conclusions drawn from the study are as follows:

Feature engineering, especially the calculation of rolling statistics and technical indicators, plays a crucial role in improving model performance.

Incorporating macroeconomic data such as news sentiment, CPI, and currency exchange rates enhances predictive accuracy.

XGBoost showed better performance compared to Random Forest, making it a preferred model for financial forecasting tasks.

The methodology and models developed can serve as a foundation for more complex financial analytics systems, including real-time risk management and automated trading systems.

References

1. Gutiérrez-Fandiño, A., Noguer i Alonso, M., Kolm, P., & Armengol-Estapé, J. (2021). FinEAS: Financial Embedding Analysis of Sentiment. arXiv preprint arXiv:2111.00526
2. Wang, S., Bai, Y.X., Ji, T., Fu, K., Wang, L., & Lu, C.T. (2023). Stock Movement and Volatility Prediction from Tweets, Macroeconomic Factors and Historical Prices. arXiv preprint arXiv:2312.03758
3. Wang, S., Bai, Y.X., Fu, K., Wang, L., Lu, C.T., & Ji, T. (2023). ALERTA-Net: A Temporal Distance-Aware Recurrent Network for Stock Movement and Volatility Prediction. arXiv preprint arXiv:2310.18706
4. Gupta, I., Madan, T.K., Singh, S., & Singh, A.K. (2022). HiSA-SMFM: Historical and Sentiment Analysis based Stock Market Forecasting Model. arXiv preprint arXiv:2203.08143
5. Schumaker, R.P., & Chen, H. AZFinText: Arizona Financial Text System. AZFinText on Wikipedia
6. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 785–794.
7. Tay, F. E. H., & Cao, L. J. (2001). Application of support vector machines in financial time series forecasting. Omega, 29(4), 309–317.
8. Zhang, Y., & Wu, L. (2009). Stock Market Prediction of S&P 500 via Combination of Improved BCO Approach and BP Neural Network. Expert Systems with Applications, 36(5), 8849–8854.
9. Nabipour, M., Nayyeri, P., Jabani, H., & Mosavi, A. (2020). Deep Learning for Stock Market Prediction. arXiv preprint arXiv:2004.01497
10. Wong, A., et al. (2023). Short-Term Stock Price Forecasting using Exogenous Variables and Machine Learning Algorithms. arXiv preprint arXiv:2309.00618
11. Ramos-Pérez, E., Alonso-González, P. J., & Núñez-Velázquez, J. J. (2020). Forecasting Volatility with a Stacked Model Based on a Hybridized Artificial Neural Network. arXiv preprint arXiv:2006.16383
12. Siswara, D., Soleh, A. M., & Wigena, A. H. (2024). Classification Modeling with RNN-Based, Random Forest, and XGBoost for Imbalanced Data: A Case of Early Crash Detection in ASEAN-5 Stock Markets. arXiv preprint arXiv:2406.07888
13. Zubair, A., & Aladejare, A. (2024). The Application of Machine Learning Techniques to Predict Stock Market Crises. Journal of Risk and Financial Management, 17(12), 554.
14. Barboza, F., Kimura, H., & Altman, E. (2017). Machine Learning Models and Bankruptcy Prediction. Expert Systems with Applications, 83, 405–417.
15. Taylor, S. J., & Letham, B. (2018). Forecasting at Scale. The American Statistician, 72(1), 37–45.

16. Siami-Namini, S., Tavakoli, N., & Namin, A. S. (2018). A Comparison of ARIMA and LSTM in Forecasting Time Series. In 2018 17th IEEE International Conference on Machine Learning and Applications (ICMLA), pp. 1394–1401.
17. Sagheer, A., & Kotb, M. (2019). Time Series Forecasting of Petroleum Production Using Deep LSTM Recurrent Networks. Neurocomputing, 323, 203–213.
18. Zhang, G. P. (2003). Time Series Forecasting Using a Hybrid ARIMA and Neural Network Model. Neurocomputing, 50, 159–175.
19. Zhang, G. P., & Qi, M. (2005). Neural Network Forecasting for Seasonal and Trend Time Series. European Journal of Operational Research, 160(2), 501–514.
20. Kim, J. (2003). Financial Time Series Forecasting Using Support Vector Machines. Neurocomputing, 55(1–2), 307–319.
21. Mahajan, V., Thakan, S., & Malik, A. Modeling and Forecasting the Volatility of NIFTY 50 Using GARCH and RNN Models.
22. Yahoo Finance - https://finance.yahoo.com/
23. News API - https://newsapi.org/
24. Reserve Bank of India (RBI) - https://www.rbi.org.in/